

SCHOOL OF ENGINEERING SECOND INTERNAL EXAMINATION, FEBRUARY 2023

19-205-0305: METALLURGY AND MATERIAL SCIENCE

Maximum
Marks: 50

Time: 2 Hrs

ANSWER ALL QUESTIONS

Questions

		Marks	CO	BL
I.	(a) Draw the steel region of the Iron -Iron Carbide phase diagram and make neat sketches of microstructures expected for four compositions between 0.0 and 1.2% C.	7	CO2	L4
	(b) Draw the T-T-T diagram for a eutectoid steel and show by sketches the microstructure you would expect for different rates of cooling.	7	CO3	L3
II.	(a) Explain about anelastic behaviour with relaxation process	7	CO4	L2
	(b) Distinguish between hardness and hardenability of steel, outlining the factors that influence each of these. Briefly explain a method to measure the hardenability of steel.	7	CO3	L2
III.	(a) Explain about critical resolved shear stress and derive the expression.	7	CO4	L3
	(b) Differentiate between ductile and brittle fracture and explain ductile brittle transition.	5	CO3	L3
	(c) Explain about strengthening mechanism and its importance.	10	CO3	L3

